Day 18 — Message Status

1. Two Different "Statuses" to Remember

When we talk about status in CPI, we must keep **two** different things apart:

#	Status Of	Question It Answers	Where You See It
①	**iFlow** (the artifact)	Is the iFlow deployed — yes or no? (*iFlow status*)	Monitor → **Manage Integration Content**
②	**Message**	What happened to one particular message that was processed?	Monitor → **Monitor Message Processing**

2. iFlow Status — Manage Integration Content (4 Types)

Status	Meaning
Started	The iFlow is deployed and running — ready to process messages
Error	Deployment or runtime start-up failed
Starting	The iFlow is **yet to get deployed** — deployment is in progress
Stopping	The iFlow is being **undeployed**

3. Design Time vs. Runtime — and Nodes

Design Time	Runtime
Where you **actually develop** the iFlow (the Design tab / canvas)	Where the process is **actually running** (deployed iFlows processing real messages)

> **Analogy from class — Bike: Manufacturing** the bike = **Design time**. The **Showroom** selling the bike = **Runtime** (also called **worker mode**).

- **Every iFlow is an artifact in CPI.**
- So in CPI, the **Monitor tab works on the Runtime node** (this ties back to Day 8: Design node = canvas editor; Runtime node = Monitor).

How Does CPI Handle Traffic? — Routes Analogy

Hyderabad	Route 1 →	**Bangalore**
	Route 2 →	
	Route 3 →	
	Route 4 →	

Scenario: only **4 routes** are available, but **10 consumers** want to travel. **What will happen?** Think of the **threads concept** — **worker nodes**: **based on the availability of nodes, the process will happen.**

- In CPI **you cannot tell how many nodes are available for your tenant.**
- **Then who manages the traffic?** In SAP CPI, the **worker nodes will process** the messages.
- SAP CPI uses a **multi-tenant architecture** (many customers' tenants share the platform, each isolated).

Four Terms Covered So Far

#	Term
①	**Design-time artifacts**
②	**Runtime artifacts**

③	**Runtime nodes**
④	**Worker nodes**

Remember especially: Runtime node and **Worker node**.

> **Added context:** the number of worker nodes is managed and scaled by SAP behind the scenes — as a developer you can't see or set it. That is exactly why the "Discarded" status below can occur.

4. Message Status — The Main Topic

Whenever we process a message, we need to know **what happened to that message** — that is the **message status**.

> **Real-time scenario:** if we work in a **support team** as an SAP CPI consultant, that person is responsible for getting the **status of all triggered messages** and acting on failures.

The 8 Message Statuses

#	Status	Meaning (from class)	Official SAP Definition (verified)
①	**Completed**	Message successfully processed to the target system	Message has been delivered to the receiver successfully
②	**Failed**	Message failed — it did not reach the target, or it stopped in the CPI flow (an Exception Subprocess ending with an **Error End Event** gives this status)	Processing failed, message not delivered, no retries possible — **a final status**
③	**Processing**	Large payload → processing takes time → status stays "Processing" meanwhile	Message is currently being processed
④	**Abandoned**	The message was processing and the user **deployed or undeployed** the same iFlow	Processing was interrupted, or the log wasn't updated for an unusually long time
⑤	**Escalated**	The flow ended at an **Escalation End Event** (Day 16)	An error occurred during processing and no retry was triggered
⑥	**Discarded**	Message already processed by another worker node, so this copy isn't processed (see section 5)	Processing discarded because the triggering timer event was already handled by another process
⑦	**Retry**	A failed message is automatically re-sent (see section 6)	An error occurred and a retry was automatically started
⑧	**Cancelled**	Message in Retry state in a JMS queue was forcefully deleted	Manual cancellation of entries in the JMS queue

> **All 8 statuses in your notes match SAP's official list exactly** — nothing missing, nothing extra. Learn them as a set; "name the message statuses in CPI" is a very common interview question.

> **Small refinements:**
>
> - **Abandoned** is broader than deploy/undeploy — SAP describes it as processing that was **interrupted** (e.g., a worker node restart or crash) or a log not updated for an unusually long time. A message stuck in Processing can turn Abandoned after an inactivity timeout (SAP's KBA mentions 24 hours). Your deploy/undeploy example is one valid trigger.
> - **Cancelled:** you wrote "forcefully deleted the queue" — SAP's wording is cancelling the **entries (messages) in the JMS queue** manually, not necessarily deleting the whole queue.
> - **Failed vs. Completed in exception handling:** an Exception Subprocess that ends with **End Message** gives **Completed**; with **Error End** gives **Failed** (Days 15–16).

5. Discarded Status — Important Point

The question: source → message → CPI (load balancer) → **worker node**. **But if we use a Timer event — who will process the message?**

- If we are **not** using a **Start Message** event but a **Timer event** instead, then in the **backend a worker node**

picks it up.

- If **two worker nodes pick the same trigger at the same time**, we would get a **duplicate record.**
- **To avoid this:** suppose we have **4 worker nodes** — worker node ① has already started processing the message; so in the **remaining 3 worker nodes**, the status of that message is shown as **Discarded.**

> **Simply: Discarded means** the message is already processed by another worker node — **that's why this message isn't processed again.** Whenever you use a Timer event, you may see Discarded entries — **this is normal and healthy**, not an error.

> **Verified:** SAP's Knowledge Base describes Discarded as: *the message processing was discarded because the triggering timer event was already handled by another process* — exactly your explanation. (Connects to Day 17: timer iFlows.)

6. Retry

What is Retry? If a message has failed and you want to **resend that message** — that is Retry.

In CPI, by default, we don't have a retry mechanism for ordinary synchronous flows. So what do we do? Use the **queue concept**:

Queue Option	Note
Solace queues	The messaging technology behind SAP Event Mesh (read via the AMQP adapter)
Kafka	Kafka-based messaging
JMS	**Java Messaging Service** queues — they do an **automatic retry** (class had a demo)

> **Nuance (verified):** "no retry by default" is true for plain HTTP/synchronous scenarios — the sender must resend. But when messages come through a **queue-based** path (JMS, or AMQP from Event Mesh), CPI **does** retry automatically and shows the status **Retry** while it does.

Interview Question — JMS Adapter

> **When you are using the JMS adapter, you must and should send the body (remember)** — otherwise you will get an exception: **InputStream with a null value.**

> **Honest note:** I couldn't confirm that exact error text in SAP's documentation, so treat the wording as your trainer's. The practical rule is sound: **make sure the message body is present (not empty and not already consumed) at the step that writes to the JMS queue** — test it in your trial tenant and watch the Trace to see the failure first-hand.

7. Cancelled

Cancelled: if the message is in **Retry state in queues (JMS)** and we **forcefully delete** the entry, then the status of that message will be **Cancelled.**

8. Quick Interview Answers

Question	Short Answer
Name the message statuses in CPI	Completed, Failed, Processing, Retry, Escalated, Abandoned, Discarded, Cancelled
What are the iFlow (artifact) statuses?	Started, Starting, Stopping, Error
What is Discarded?	A duplicate trigger (e.g., timer) already handled by another worker node — not an error
Is Failed final?	Yes — no more retries are possible
How do you get automatic retry?	Use queues — JMS (or AMQP/Event Mesh); plain sync flows have no automatic retry
What happens when a message in a JMS retry loop is deleted manually?	Status becomes **Cancelled**

Where do you monitor messages vs. iFlows?	Messages: Monitor Message Processing. iFlows: Manage Integration Content